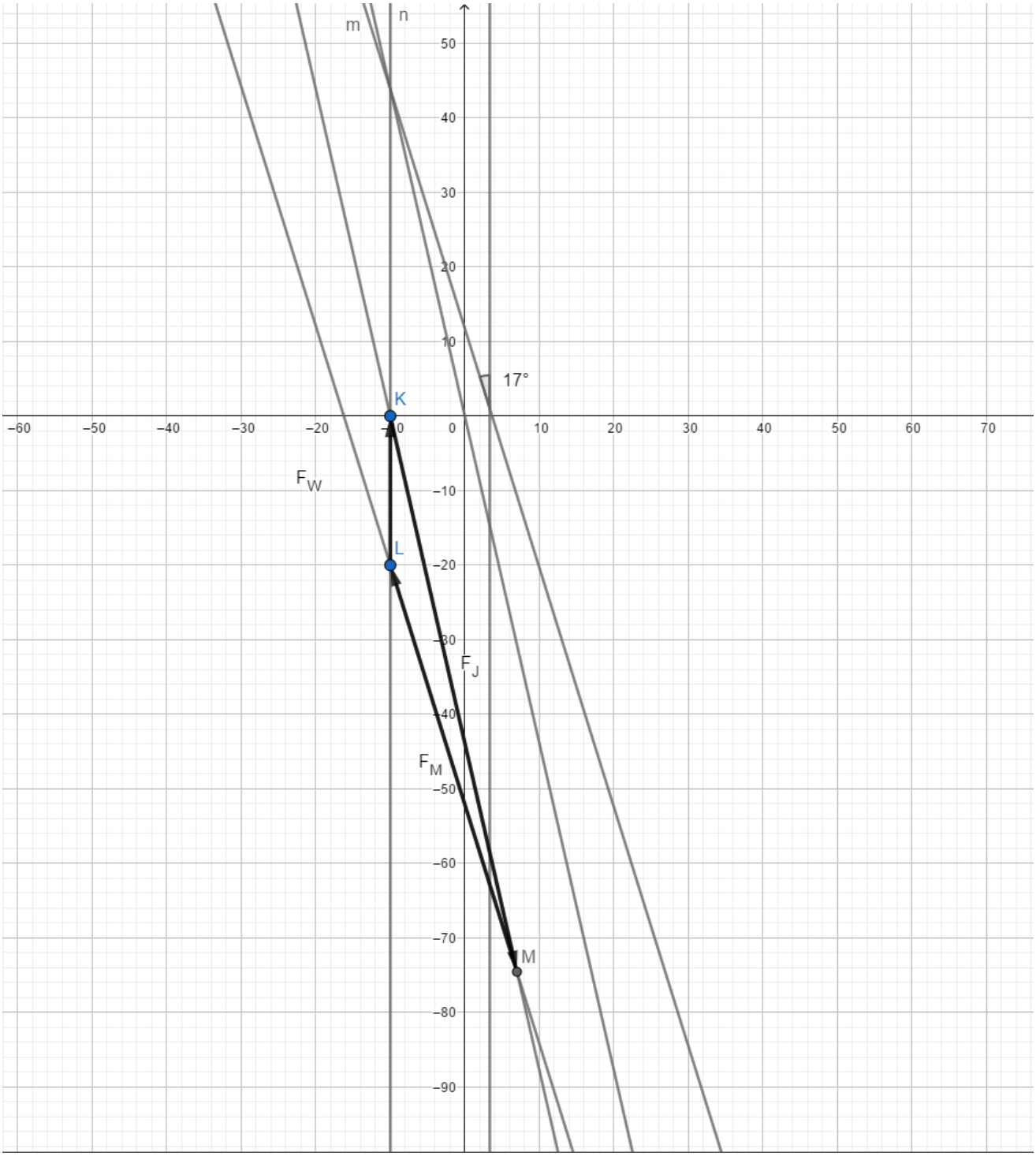
Rovnováha

a)

$$\vec{W}; \vec{W}[0; 800N]; \overrightarrow{r_w}[0, -10cm]$$
$$\overrightarrow{F_M}; \overrightarrow{F_M}[-(F_M * cos15^\circ); F_M * sin15^\circ]; \overrightarrow{r_M}[3.5cm * cos17^\circ; 3.5cm * sin17^\circ]$$
$$\overrightarrow{F_J}; \overrightarrow{F_J}[F_{Jx}; F_{Jy}]; \overrightarrow{r_J}[0, 0]$$

$$--\times--$$
$$0 - (F_M * cos15^\circ) + F_{Jx} = 0$$
$$800N + F_M * sin15^\circ + F_{Jy} = 0$$
$$F_M * sin15^\circ * \overrightarrow{r_M} - 800N * \overrightarrow{r_W} = 0$$
$$----$$
$$F_{Jx} = F_M * cos15^\circ$$
$$F_{Jy} = -(800N + F_M * sin15^\circ)$$
$$F_M = \frac{800N * \overrightarrow{r_W}}{sin15^\circ * \overrightarrow{r_M}}$$
$$----$$
$$F_M = \frac{800N * -10cm}{sin15^\circ * 3.5cm * cos17^\circ}$$
$$F_M = \frac{-8000}{3.5 * \frac{sin(15+17)+sin(15-17)}{2}}$$
$$F_M = \frac{-8000}{3.5 * \frac{sin(32)+sin(-2)}{2}}$$
$$F_M = \frac{-8000}{3.5 * \frac{0.495}{2}}$$
$$F_M = \frac{-8000}{3.5 * \frac{99}{400}}$$
$$F_M = \frac{-8000}{\frac{693}{800}}$$
$$F_M = \mathbf{-9235.209N}$$
$$----$$
$$F_{Jx} = -9235.209N * cos15^\circ$$
$$F_{Jx} = \mathbf{-8920.527N}$$
$$----$$
$$F_{Jy} = -(800N + (-9235.209N) * sin15^\circ)$$
$$F_{Jy} = -(800N - 2390.248)$$
$$F_{Jy} = \mathbf{1590.248N}$$

b)



c)

Nevýhoda -> je potřeba větší síla ke stejnému pohybu

Výhoda ->není potřeba tak velká noha, která bý následně tvarem stěžovala pohyb